

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0541

Roanoke Rapids 138 kV
August 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0541. The Interconnection Customer proposes a 100.7 MW generating facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on November 21, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 219 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$319,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yadkin-No Mans Land 138 KV	Mickens-Miravista 345 KV	109.0	468	6.2
Mickens-Mira Vista 345 KV line	Mccartney-N Tower Tap 345 KV	108.8	1,177	16.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yadkin-No Mans Land 138 KV	Reconductor line	\$871,000	1% (\$9,000)
Mickens-Mira Vista 345 KV line	Reconductor line	\$17,381,000	2% (\$310,000)
	Total allocated to project		\$319,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.